2.6 The Arithmetic–Geometric Mean

WORKSHEET

The question

In this question, the following theorem may be used without proof.

Let $u_1, u_2, \ldots$ be a sequence of real numbers. If the sequence is

- *bounded above, so $u_n \leq b$ for all n, where b is some fixed number*
- *and increasing, so $u_n \leq u_{n+1}$ for all n*

then there is a number $L \leq b$ such that $u_n \to L$ as $n \to \infty$.

For positive real numbers x and y, define $\mathrm{a}(x,y) = \frac{1}{2}(x+y)$ and $\mathrm{g}(x,y) = \sqrt{xy}$. Let x_0 and y_0 be two positive real numbers with $y_0 < x_0$ and define, for $n \geq 0$,

$$x_{n+1} = \mathrm{a}(x_n, y_n), \qquad y_{n+1} = \mathrm{g}(x_n, y_n).$$

(i) By considering $\left(\sqrt{x_n} - \sqrt{y_n}\right)^2$, show that $y_{n+1} < x_{n+1}$, for $n \geq 0$. Show further that, for $n \geq 0$,

- $x_{n+1} < x_n$
- $y_n < y_{n+1}$.

Deduce that there is a value M such that $y_n \to M$ as $n \to \infty$.

Show that $0 < x_{n+1} - y_{n+1} < \frac{1}{2}(x_n - y_n)$ and hence that $x_n - y_n \to 0$ as $n \to \infty$.

Explain why x_n also tends to M as $n \to \infty$.

(ii) Let

$$\mathrm{I}(p,q) = \int_0^\infty \frac{1}{\sqrt{(p^2+x^2)(q^2+x^2)}}\,\mathrm{d}x,$$

where p and q are positive real numbers with $q < p$.

Show, using the substitution $t = \frac{1}{2}\left(x - \frac{pq}{x}\right)$ in the integral

$$\int_{-\infty}^\infty \frac{1}{\sqrt{\left(\frac{1}{4}(p+q)^2 + t^2\right)(pq+t^2)}}\,\mathrm{d}t,$$

that

$$\mathrm{I}(p,q) = \mathrm{I}\big(\mathrm{a}(p,q), \mathrm{g}(p,q)\big).$$

Hence evaluate $\mathrm{I}(x_0, y_0)$ in terms of M.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Why is the arithmetic mean of two different positive numbers always bigger than their geometric mean? Can you prove it in one line?

STAGE 1 The two sequences squeeze together

Ask yourself Why is the arithmetic mean of x_n and y_n less than x_n, and the geometric mean bigger than y_n?

STAGE 2 The substitution

Ask yourself With $t = \frac{1}{2}\left(x - \frac{pq}{x}\right)$, compute $t^2 + pq$. Then compute $t^2 + \frac{1}{4}(p+q)^2$. Do either of them factorise nicely?

Pause and try Put these together. As x runs from 0 to ∞, what does t do? What does the t-integral become?

STAGE 3 Evaluating the integral

Ask yourself Apply this identity repeatedly, starting from (x_0, y_0). What is $\mathrm{I}(x_0, y_0)$ equal to after n steps? What happens as $n \to \infty$?

Try this next

1. **Quadratic convergence.** Prove the identity $x_{n+1} - y_{n+1} = \dfrac{(x_n - y_n)^2}{2(\sqrt{x_n} + \sqrt{y_n})^2}$, and deduce that $x_{n+1} - y_{n+1} \le \dfrac{(x_n - y_n)^2}{8y_0}$.

2. **By hand.** Compute $M(2,1)$ to six decimal places using a calculator. How many steps did you need?

3. **Scaling.** Show that $M(\lambda x, \lambda y) = \lambda M(x,y)$ for $\lambda > 0$, and that $\mathrm{I}(\lambda p, \lambda q) = \lambda^{-1}\mathrm{I}(p,q)$. Check that these two facts are consistent with $\mathrm{I}(x_0, y_0) = \pi/(2M)$.

4. **The pendulum.** Use the formula above to find, to three significant figures, how much longer the period of a pendulum released from 60° is than the small-angle prediction.

Further reading

- **David Cox, "The arithmetic–geometric mean of Gauss"**, *L'Enseignement Mathématique* 30 (1984). The classic account of the history and mathematics; the first sections need only calculus.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The definitive book on the AGM and fast computation of π.
- **Jonathan Borwein & Keith Devlin, *The Computer as Crucible*** (A. K. Peters). Experimental mathematics, including discoveries like Gauss's, made by spotting numerical coincidences.

The full guided solution, and what the question leads to at university, is in Section 2.6:
https://beyondstep.keithmcnulty.org/ch02/agm.html